

4.8.7 Project Design - Room Allocation Engine

4.8.7.1 Use Case Diagram

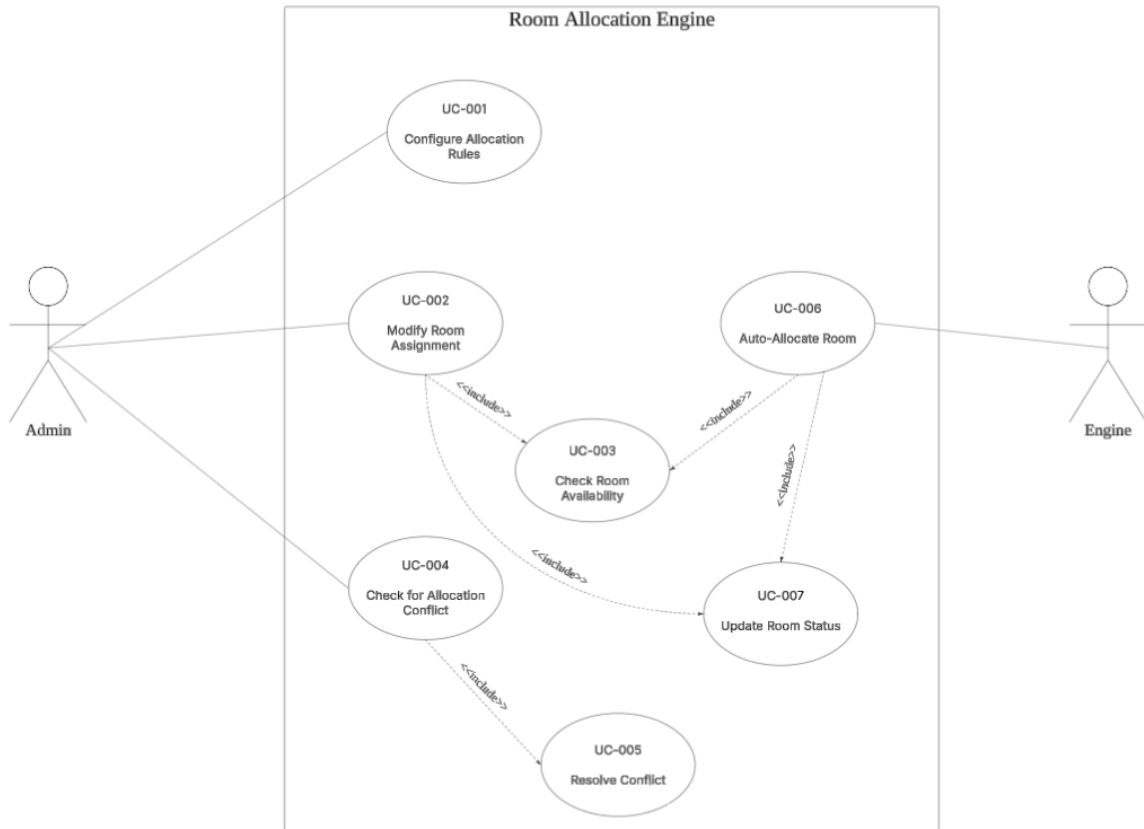


Figure 1.5: Use Case Diagram for Room Allocation Engine

Source:

https://lucid.app/lucidchart/b7f3b5d3-a742-410b-a4c5-38dfa744d49f/edit?invitationId=inv_e2a734bd-f000-4218-99ca-bc6f46c5ac4f

4.8.7.2 Use Case Description

4.8.7.2.1 UC-001 Configure Allocation Rules

Item	Description
Use Case ID	UC-001
Use Case Name	Configure Allocation Rules
Actor	Admin
Description	Allows admin to define and manage room allocation rules such as gender, year of study and room capacity
Precondition	Admin is authenticated and has access rights
Trigger	Admin select the option to configure allocation rules
Main Flow	1. Admin accesses allocation rule option 2. Admin defines allocation rules 3. Admin save the rules 4. System validates and store the rules
Alternate Flow	1. Invalid rule configuration 2. System displays an error message
Postcondition	Allocation rules are successfully saved and ready for use by the engine

4.8.7.2.2 UC-002 Modify Room Assignment

Item	Description
Use Case ID	UC-002
Use Case Name	Modify Room Assignment
Actor	Admin
Description	Allow admin to manually assign or change a student's room allocation
Precondition	Admin is authenticated and student record exists
Trigger	Admin select the option to modify room

	assignment
Main Flow	1. Admin search for student record 2. System display the student information 3. Admin select a room for the student 4. System checks room availability (UC-003) 5. System update the room assignment
Alternate Flow	1. Selected room unavailable 2. System prompts admin to select another room
Postcondition	Student's room assignment is updated successfully

4.8.7.2.3 UC-003 Check Room Availability

Item	Description
Use Case ID	UC-003
Use Case Name	Check Room Availability
Actor	Admin & Engine
Description	Check the status of room availability before assignment or modification
Precondition	Room data exists in the system
Trigger	From UC-002 & UC-006
Main Flow	1. System retrieves room data 2. System returns availability result 3. Admin & Engine identifies vacant and occupied rooms
Alternate Flow	1. Room data not available 2. System displays an error message
Postcondition	Room availability status is provided to the calling use cases

4.8.7.2.4 UC-004 Check for Allocation Conflict

Item	Description
Use Case ID	UC-004
Use Case Name	Check for Allocation Conflict
Actor	Admin
Description	Allows the admin to check for conflicts such as duplicate room assignments or overcapacity issues
Precondition	Student room allocation data exists
Trigger	Admin selects the option to see student allocation data and check for allocation conflict
Main Flow	1. Admin accesses the student room allocation data 2. System displays the list of data 3. Admin initiates conflict check 4. System scans room allocation records 5. System identifies conflicting allocations
Alternate Flow	1. No conflict detected 2. System displays confirmation message
Postcondition	Allocation conflicts are identified and displayed

4.8.7.2.5 UC-005 Resolve Conflict

Item	Description
Use Case ID	UC-005
Use Case Name	Resolve Conflict
Actor	Admin
Description	Allows admin to resolve allocation conflicts
Precondition	Allocation conflict exists
Trigger	Admin select a conflict record to resolve

Main Flow	1. System displays conflict details 2. Admin reassign room 3. System confirms resolution
Postcondition	Allocation conflict is resolved and records are updated

4.8.7.2.6 UC-006 Auto-allocate Room

Item	Description
Use Case ID	UC-006
Use Case Name	Auto-allocate Room
Actor	Engine
Description	Automatically allocates rooms to student based on predefined rules
Precondition	Allocation rules exist and student data is available
Trigger	Engine initiates auto-allocation process
Main Flow	1. Engine retrieves student data 2. Engine check room availability (UC-003) 3. Engine use allocation rules 4. Engine assign rooms to students 5. Engine updates room status (UC-007)
Alternate Flow	1. No suitable room found 2. Allocation are marked as conflict for further action
Postcondition	Rooms are allocated successfully or conflicts are flagged

4.8.7.2.7 UC-007 Update Room Status

Item	Description
Use Case ID	UC-007
Use Case Name	Update Room Status

Actor	Admin & Engine
Description	Updates the occupancy status of rooms after allocation or modification
Precondition	Room assignment has been performed
Trigger	Included by UC-002 & UC-006
Main Flow	1. System updates room occupancy status 2. System saves updated room data
Postcondition	Room status saved successfully

4.8.7.3 Activity Diagram

4.8.7.3.1 Configure Allocation Rules (UC-001)

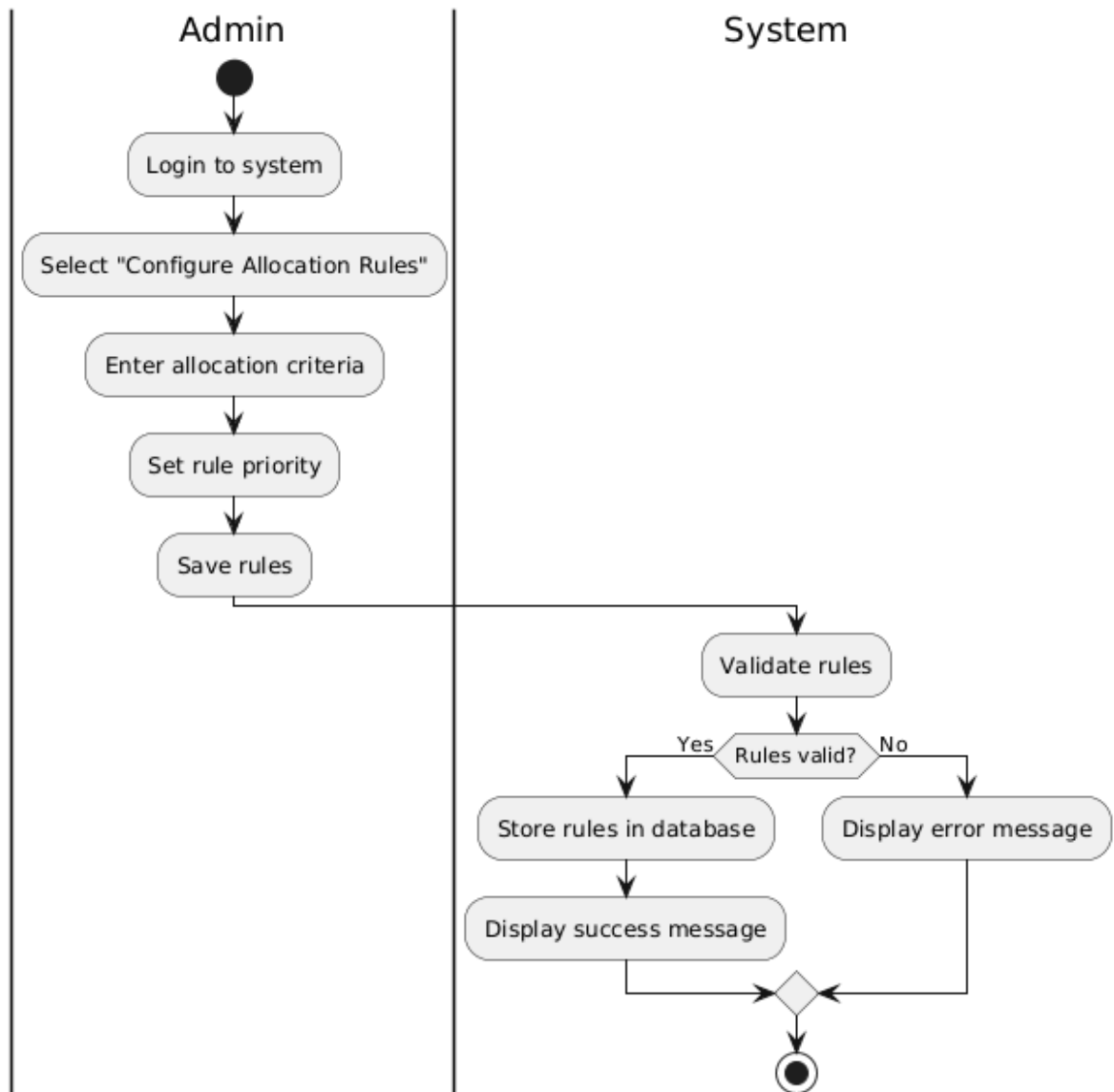


Figure 1.6: Activity Diagram for Configure Allocation Rules

4.8.7.3.2 Modify Room Assignment (UC-002)

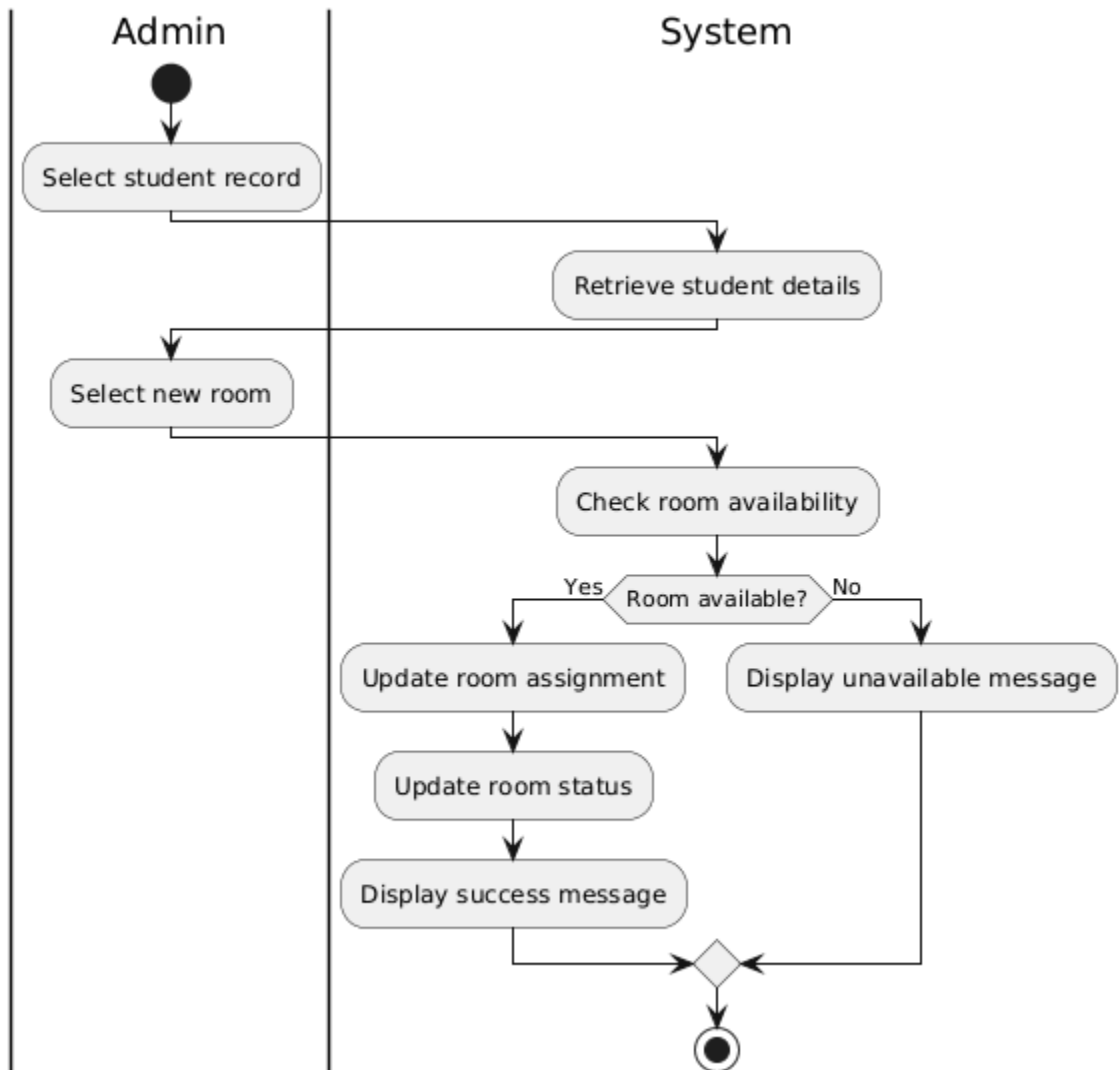


Figure 1.7: Activity Diagram for Modify Room Assignment

4.8.7.3.3 Check Room Availability (UC-003)

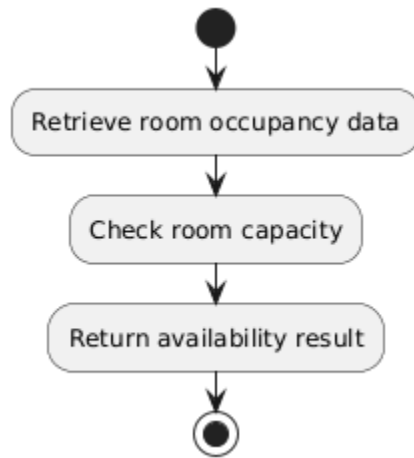


Figure 1.8: Activity Diagram for Check Room Availability

4.8.7.3.4 Check for Allocation Conflict & Resolve Conflict (UC-004 & UC-005)

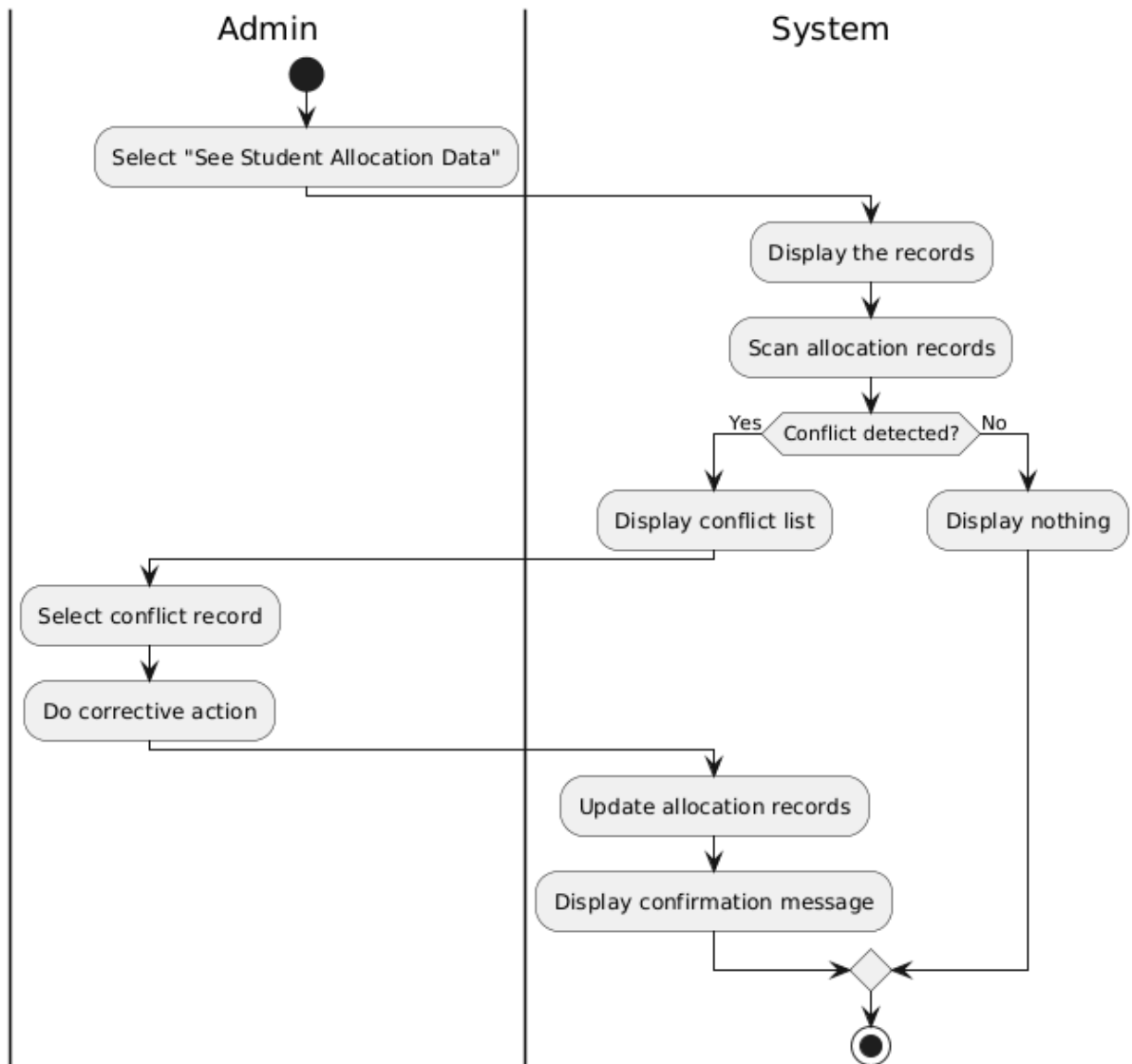


Figure 1.9: Activity Diagram for Check for Allocation & Resolve Conflict

4.8.7.3.5 Auto-Allocate Room (UC-006)

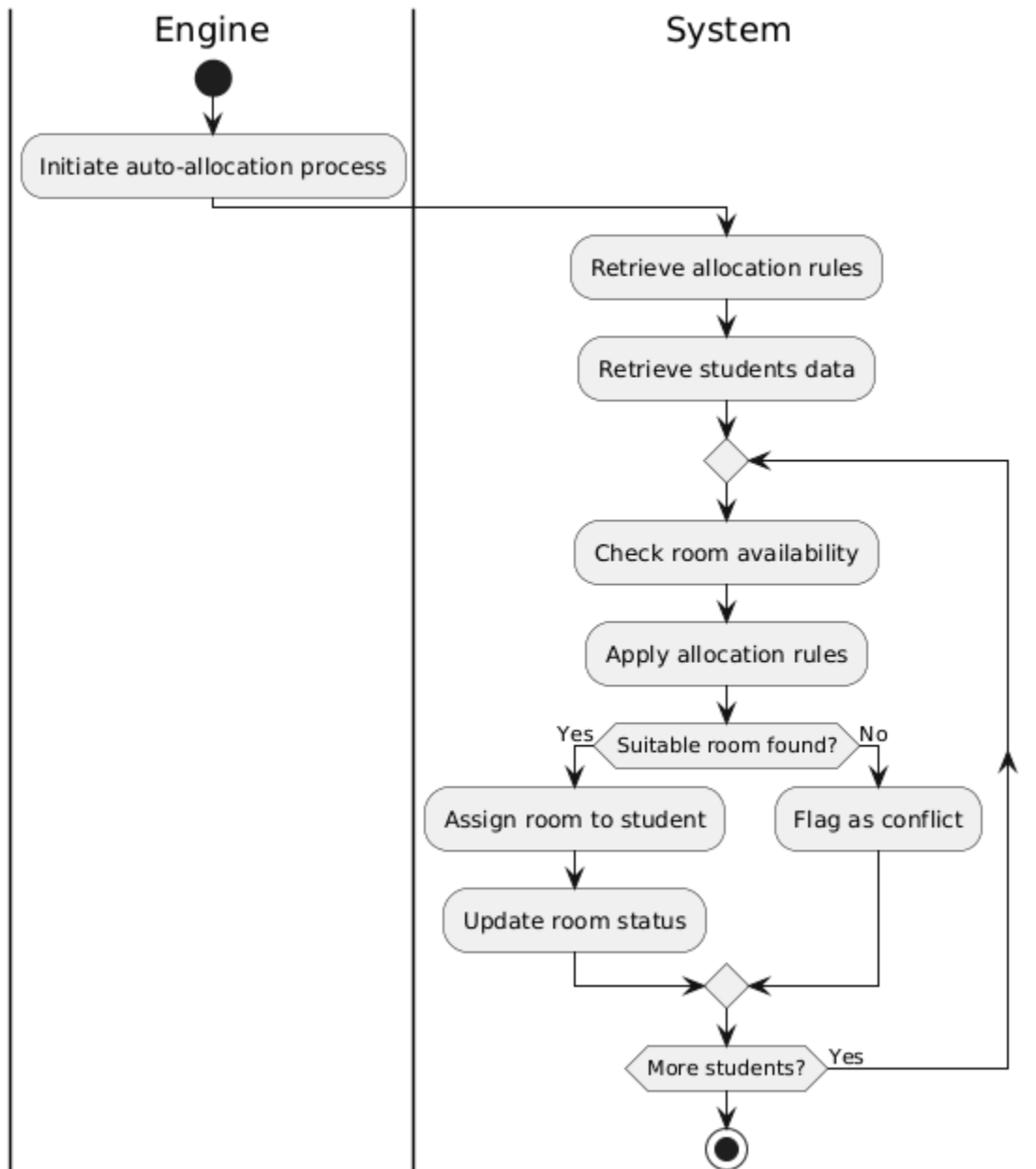


Figure 1.10: Activity Diagram for Auto-Allocate Room

4.8.7.4 Sequence Diagram

4.8.7.4.1 Configure Allocation Rules (UC-001)

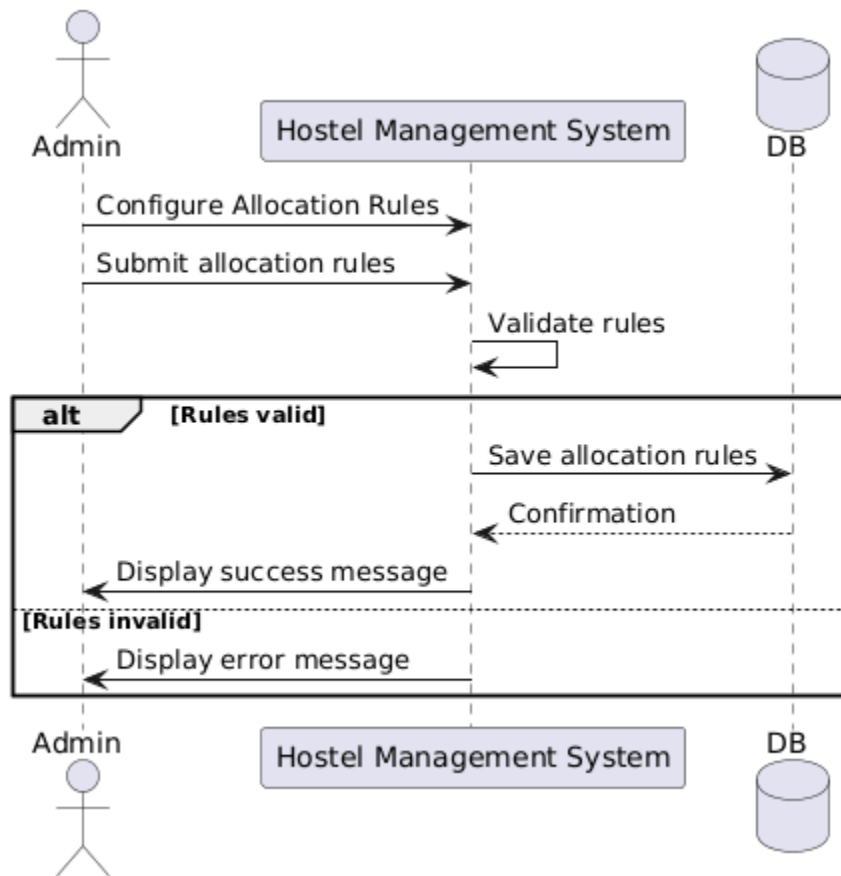


Figure 1.11: Sequence Diagram for Configure Allocation Rules

4.8.7.4.2 Modify Room Assignment (UC-002)

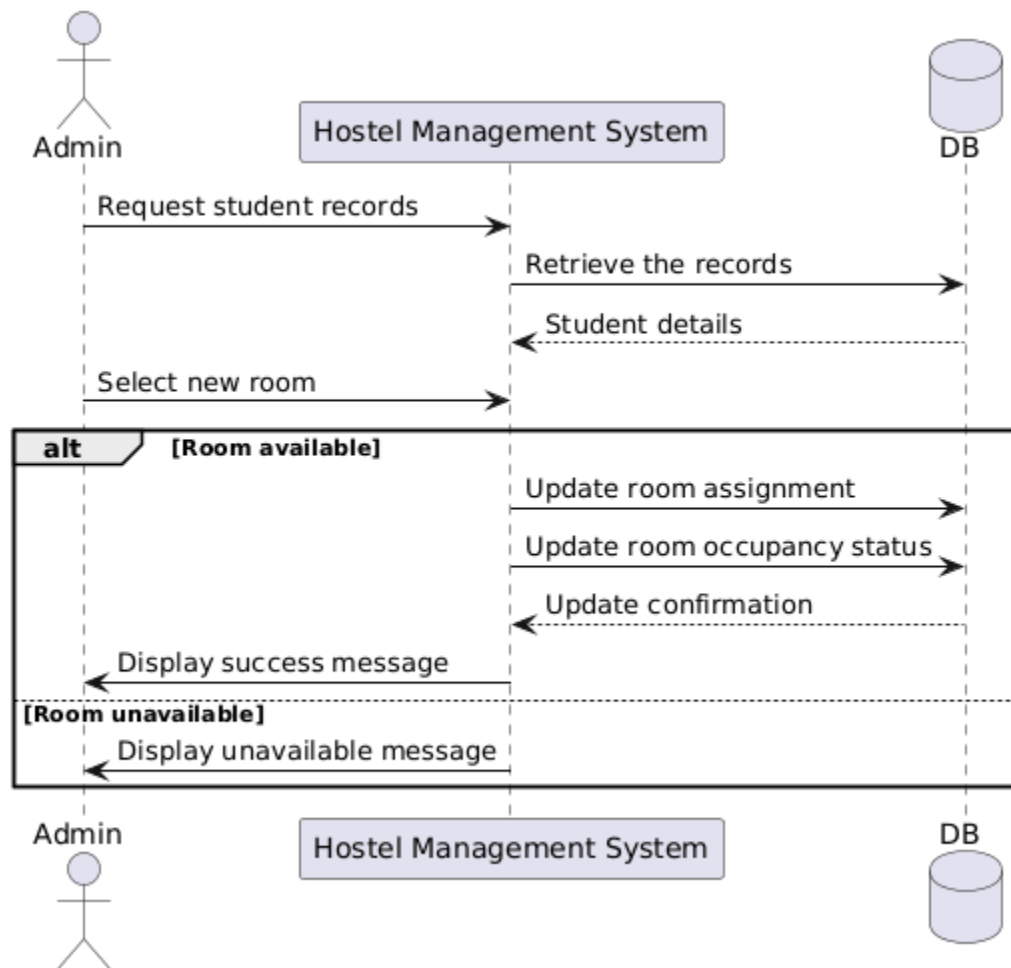


Figure 1.12: Sequence Diagram for Modify Room Assignment

4.8.7.4.3 Check for Allocation Conflict & Resolve Conflict (UC-004 & UC-005)

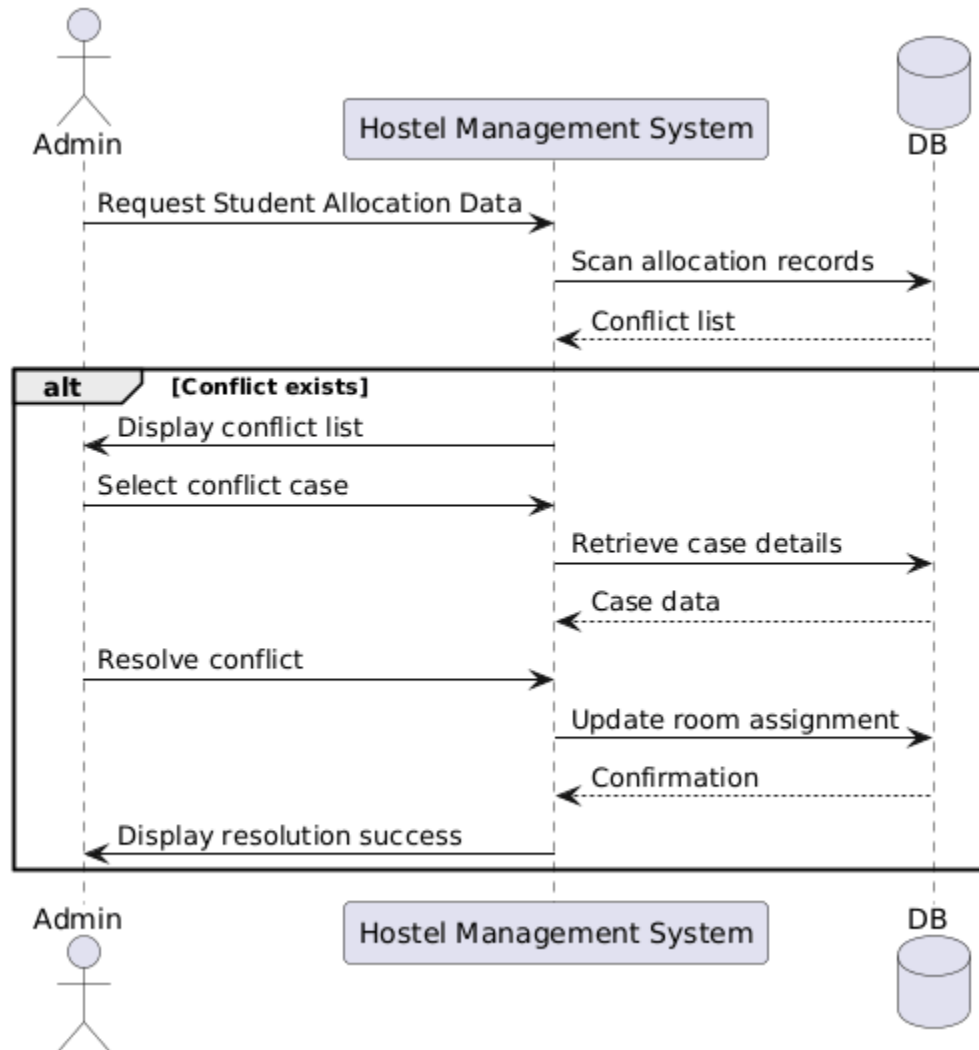


Figure 1.13: Sequence Diagram for Check for Allocation Conflict & Resolve Conflict

4.8.7.4.4 Auto-Allocate Room (UC-006)

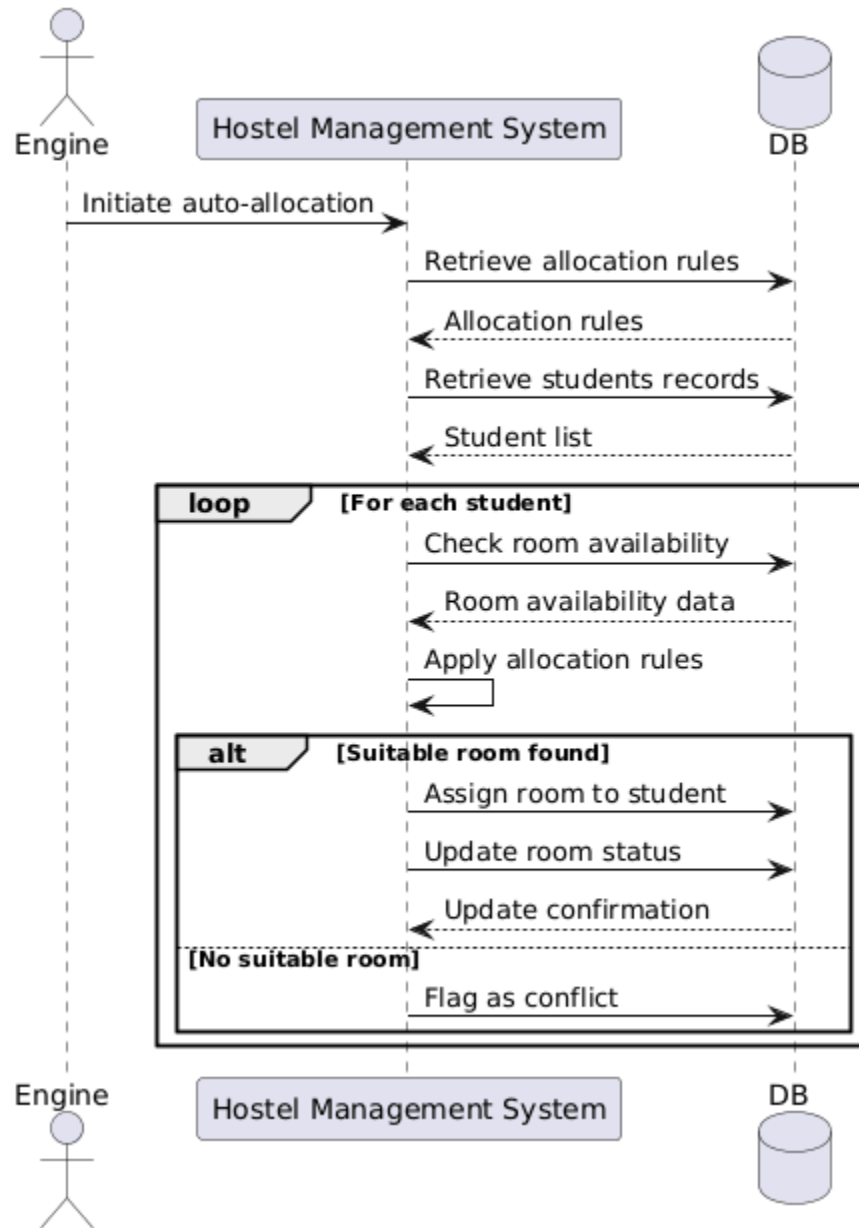


Figure 1.14: Sequence Diagram for Auto-Allocate Room